Original-Row Proximal Splitting for Side-Specific and Directional Peak Penalties: Two Negative Timing Studies

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Abstract

Original-row proximal updates can represent selected soft interval penalties without adding explicit departure variables to a quadratic program's principal system. We derive a side-specific separable linear–quadratic update and a directional disjoint group-peak update in a fixed positive diagonal metric. The grouped construction follows a box-residual identity under coordinate contractions and reduces its proximal operation to a weighted scalar threshold. Exact lifting, directional row splitting and compact epigraphs already express the same objectives; no expressibility or algorithmic-priority claim is made. We evaluate the two implementations in separate registered synthetic studies. The first contains 456 instances and 168 held-out accepted pairs: median relative cold end-to-end reduction 0.499839%, with 95% seed-cluster limits -3.291967% and 3.944835%. The second contains 210 repeated-forecast instances, with a 144-member held-out primary score of -212.449875%, limits -225.336618% and -192.873759%, and 9 retained failed pairs. Neither predicted ten-percent timing improvement is established. Original-space objective, hard-row and dual checks distinguish mathematical equivalence from computational cost. Matched formulation, conditioning and workspace comparisons quantify descriptive contrasts without asserting an unperformed causal ablation. The results show that fewer principal variables or rows do not by themselves establish a faster complete optimization operation on these tested populations.

1 Introduction

Soft interval constraints specify a price for violating a preference while leaving selected physical limits immutable. In predictive control, a preferred state range and an actuator capacity are not interchangeable constraints. In energy scheduling, an upper grid departure and a lower comfort departure can have different units and consequences. The numerical problem must state which rows may yield and which objective measures that choice before a solver comparison has a definite meaning. Convex quadratic objectives with such interval penalties provide a useful, transparent setting for that analysis.

An ordinary slack lift expresses many of these prices exactly, but enlarges the mathematical system. Generalized slack projections replace selected slack variables with proximal row operations and can preserve the original principal dimensions [33, 20]. That observation motivates a computational hypothesis: an original-row implementation might reduce the cost of a complete optimization operation. It does not prove the hypothesis. A proximal operation can itself be expensive, sparse fill depends on more than dimension, and preparation, validation and recovery contribute to the measured operation.

We study two related original-row updates. The first applies separate nonnegative linear–quadratic prices to lower and upper interval departures. Its scalar proximal minimizer follows from exterior quadratic stationarity and endpoint subgradient conditions. The second prices directional disjoint group peaks. A box-residual identity under coordinate contractions reduces its positive diagonal-metric proximal operation to a weighted scalar threshold. We state the precise fixed-metric and relaxation scope, preserve hard rows separately, and test outputs in original physical coordinates. Existing side-weighted active-set formulations, directional row splitting and compact epigraphs already express the same objectives [1, 33]. Consequently, neither unequal side costs nor directional expressibility is a priority claim.

The two computational evaluations are separately registered synthetic studies. The original study contains 456 instances, with 168 primary paired observations from 24 distinct held-out problems and seven repetitions across six seed clusters. Its median cold end-to-end reduction is 0.499839%, with 95% limits -3.291967% and 3.944835%. The grouped study contains 210 instances and a complete 144-instance held-out population. Its primary score is -212.449875%, with limits -225.336618% and -192.873759%, including 9 failed pairs. Both predicted 10% improvements are unestablished. These findings concern different measurement boundaries and cannot be pooled into a new favorable estimator.

The contribution is a derivation and implementation analysis with explicit negative computational evidence. We provide the side-specific proximal formula, the contraction-based box reduction, the directional threshold and conjugate, and the fixed-metric splitting conditions. We compare exact objective-matched references and retain compatibility limits, unsuccessful observations and original-space acceptance. Component comparisons examine formulation choice, conditioning and workspace reuse while stating where no causal intervention was performed. The results establish mathematical equivalence and tested implementation behavior within their stated domains, not a uniform runtime advantage, operational safety or a new expressibility theorem beyond the identified prior art.

The following sections position the methods within convex proximal and quadratic-programming literature, define the model, derive each proposed update separately and state the theory's limits. Experimental design and results keep the two populations and confidence procedures distinct. The ablation analysis is explicitly limited to retained component comparisons; discussion and limitations explain why correct original-row updates need not improve complete-operation time. Reproducibility relies on the exact archived inputs and outputs rather than on post-hoc reconstruction of a successful subset.

2 Related work

Convex analysis supplies the modeling and duality framework rather than a claim of new mathematics from changing a solver interface. Closed convex penalties, support functions and conjugates determine whether a recovered dual is consistent with an original objective [30, 4].

Moreau's proximal decomposition and proximal-point methods explain why strongly convex auxiliary minimizations can be useful without changing the target objective [23, 31]. Parikh and Boyd describe proximal operators and their composition limits in optimization [27]. These sources support the vocabulary and identities used below; they do not assert that an arbitrary composition of projection and a proximal map is valid for every convex penalty.

Douglas–Rachford and ADMM splitting have a substantial convergence literature. Lions and Mercier analyze splitting for sums of monotone operators, while Eckstein and Bertsekas establish the relation to the proximal-point framework [21, 11]. The distributed-optimization treatment of ADMM makes the role of primal feasibility and dual consistency explicit [3]. Fixed-metric rate analyses require quantitative assumptions beyond basic convexity [14]. We use these results to delimit a fixed positive metric and a fixed relaxation in its admissible range. We do not infer a variable-metric theorem or universal linear rate from the existence of a closed-form row update.

OSQP combines operator splitting with a sparse quasi-definite system, warm starts and numerical infeasibility checks [34]. Lew and coauthors eliminate quadratic slack effects without increasing the main problem size; Schwan, Lew and Plancher generalize slack projection to supported convex penalty families [20, 33]. The latter work is the closest native projection reference. Its unconditional penalty assumptions include sign symmetry that is narrower than arbitrary side-specific costs. Its published grouped path nevertheless becomes a directional reference after an exact one-sided row transformation. Removing that reference would overstate the modeling difference of an original-row implementation.

Active-set approaches provide another important comparison. The dual active-set foundation of Goldfarb and Idnani and the recursive updates of DAQP address structured quadratic programs without relying on the same projection iteration [15, 1]. DAQP exposes side-specific soft weights, so asymmetry alone is established capability. Its proximal linear-programming extension also supplies a relevant registered reference for suitable grouped epigraphs [2]. qpOASES exploits parametric QP structure through an online active-set strategy [13]. These methods show why a retained active set or a vector-update interface cannot be presented as a new mechanism simply because it appears in a different software wrapper.

FlexQP and Deep FlexQP distinguish fixed elastic splitting from learned numerical-parameter selection [26]. The executable comparison here reconstructs only the fixed paper equations. ElastiQP supplies a separate published pure-linear elastic reference with hard-equality support on its admitted domain [24]. PIQP illustrates a proximal interior-point approach to quadratic programming [32]; it is numerical background, not an extra unrecorded timed competitor in the present populations. Classical linear programming and numerical optimization texts explain the dual bounds, curvature assumptions and conditioning issues needed to interpret these different formulations [35, 25].

The grouped construction also has directly relevant prior art. Orthant- monotonicity results concern coordinatewise contraction behavior and its relationship to signed convex geometry [6]. Weighted projection and threshold algorithms are established, including weighted norm-ball, simplex and linear-time projection calculations [28, 10, 7]. The scalar active-prefix equation below specializes those ideas to a directional peak and diagonal metric; sorting or thresholding by itself is not claimed as a new algorithmic primitive. Floating-point and symmetry- preserving scaling work further cautions against confusing algebraic equivalence with identical numerical cost [17, 19].

Finally, optimization benchmarking depends on explicit success criteria and problem inventories, as performance profiles and reproducible test collections make clear [9, 16]. Bootstrap

methods provide a framework for uncertainty, while clustered inference emphasizes dependence within shared experimental units [12, 8, 5]. These sources motivate complete populations, seed-level resampling and retained failures. They do not justify a post-hoc subset or guarantee precise coverage with only six seed clusters. The present experiments therefore report their negative registered endpoints separately and restrict descriptive component plots to their actual, narrower interpretation.

3 Problem formulation and notation

Let $P \succeq 0$ be symmetric, q finite, A sparse and $x \in \mathbb{R}^n$ the original decision. Each row value $z_i = A_i x$ has an interval $[\ell_i, u_i]$ whose endpoints may be explicitly absent. Hard rows form $\mathcal{H}$; all other admitted rows have a specified convex price. The model is

$$\min_x \ \tfrac{1}{2} x^\top P x + q^\top x + g(Ax) \quad \text{subject to} \quad \ell_i \le A_i x \le u_i \quad (i \in \mathcal{H}). \tag{1}$$

The soft price g is separable for the first method and a sum over disjoint directional groups for the second. Write $g_{\text{all}}(z) = g(z) + \sum_{i \in \mathcal{H}} \iota_{[\ell_i, u_i]}(z_i)$, where each indicator is zero on its hard interval and infinite outside. Both constructions keep physical hard policy distinct from numerical penalty parameters. An elastic optimum can purchase a preferred-row violation; that does not change what constitutes hard feasibility.

Lower and upper departures are $v_{L,i} = (\ell_i - z_i)_+$ and $v_{U,i} = (z_i - u_i)_+$. Corrective slack has sign $s_i = v_{L,i} - v_{U,i}$, opposite to the signed violation residual $z_i - \Pi_C(z)_i$. The distinction matters when conjugates and original row multipliers are recovered. Upper finite-side multipliers enter the original dual positively; lower finite-side multipliers enter negatively. The fixed positive diagonal proximal metric is $H = \operatorname{diag}(h_i)$, $h_i > 0$. It is not a changing physical price or an arbitrary dense coupling matrix.

Table 1: Symbols and dimensions for the two proximal constructions. Physical row and decision units belong to each original model; computational duration and model prediction time remain different quantities.

Symbol	Meaning	Dimension
$x, z = Ax$	Decision and original row value	Decision / row unit
P, q, A	Curvature, linear cost, row map	Cost/decision unit squared; cost/decision unit; row/decision unit
C, Π_C, r	Interval box, box projection and residual	Row space / row unit
a_i, b_i	Linear and quadratic side price	Cost/row unit; cost/row unit squared
$c_{U,i}, c_{L,i}$	Directional scale	Peak unit/row unit
a_g, b_g	Linear and quadratic peak price	Cost/peak unit; cost/peak unit squared
Ψ, g_C	Residual and eliminated interval penalty	Cost
H, α	Fixed proximal metric and relaxation	Cost/row unit squared; dimensionless
y, σ_C, Ψ^*	Original multiplier, support and conjugate	Cost/row unit; cost; cost
T, R	Computational duration and relative reduction	Seconds; dimensionless

Table 1 fixes the interpretation used in derivations and figures. For a directional group, scales make physically different departures comparable in one peak unit. With metres as row units and a dimensionless peak, directional scales have inverse-metre units. A reciprocal row conversion

scales both endpoints and the scale coefficient; changing only one directional coefficient instead changes the policy. We do not add raw departures with incompatible dimensions or interpret a population index as physical time.

Input validity does not imply solution existence. A hard model may be infeasible; semidefinite curvature may admit unbounded directions or nonunique optimizers. The splitting statements assume an attained saddle point in addition to their closed-convex conditions. The native grouped interface additionally requires nonempty boxes, disjoint groups and positive finite directional scales. Mathematical extension to overlapping groups, dense metrics or nonconvex prices is not inferred from tests of the admitted cases. These assumptions keep the implemented objective, the proximal proof and the experimental acceptance domain aligned.

The empirical quantities have their own dimensions: complete-call durations are seconds, relative reductions are dimensionless and displayed as percent, and counts identify observations or instances as specified by each protocol. An unavailable result carries no numerical duration or objective for inference. These conventions prevent missing values from becoming successful zero-cost operations when tables or statistical arrays are assembled.

4 Proposed method: side-specific interval updates

The first construction is a scalar interval proximal operation inserted into an original-row quadratic-programming split. Its input is not a single generic softness number. Each finite lower and upper side has its own nonnegative linear price and quadratic curvature. This permits a preferred lower departure to have a different economic or engineering meaning from an upper departure. Physical hard rows remain a separate indicator constraint. The construction is related to generalized slack projection, but its lower and upper branches are derived directly rather than justified by assuming a sign-symmetric penalty [33, 20].

For one nonempty interval $[\ell, u]$, write

$$\phi(z) = a_L(\ell - z)_+ + a_U(z - u)_+ + \tfrac{1}{2}b_L(\ell - z)_+^2 + \tfrac{1}{2}b_U(z - u)_+^2, \quad a_L, a_U, b_L, b_U \geq 0. \tag{2}$$

If a side is absent its term is omitted. A zero linear or quadratic coefficient is admissible; a negative quadratic coefficient is not. On a hard row the corresponding term of g_{all} is the interval indicator rather than approximated by a very large finite price. The distinction avoids confusing an elastic optimum with physical hard feasibility. It also keeps the mathematical domain unchanged when a local cost multiplier is varied.

For a temporary scalar v and fixed positive metric h, define

$$p_h(v) = \operatorname{argmin}_z \left\{ \phi(z) + \tfrac{h}{2}(z - v)^2 \right\}. \tag{3}$$

The displacement term is strongly convex even if both quadratic side prices vanish, so the proximal minimizer is unique. The penalty is zero on the preferred interval and nondecreasing in each exterior departure. Consequently an optimizer need not be sought beyond the segment joining v to its nearest interval endpoint. This geometric restriction is a consequence of convex monotone exterior prices, not a numerical clipping heuristic that changes an already computed solution.

If v lies inside the interval, $z = v$ simultaneously minimizes the zero penalty and the squared displacement. If $v > u$, write $d = z - u \geq 0$. The derivative on the positive exterior branch is

$$(h + b_U)d + a_U - h(v - u). \tag{4}$$

If this derivative is nonnegative at zero, the endpoint satisfies the subgradient condition. Otherwise its unique positive root gives the exterior solution. Reflecting the displacement produces the corresponding lower-side equation with a_L, b_L. The complete map is

$$p_h(v) = \begin{cases} \ell - \dfrac{[h(\ell - v) - a_L]_+}{h + b_L}, & v < \ell, \\ v, & \ell \le v \le u, \\ u + \dfrac{[h(v - u) - a_U]_+}{h + b_U}, & v > u. \end{cases} \tag{5}$$

All denominators remain positive. A hard row uses interval projection, and a free soft side returns the unconstrained temporary value on that side. At an equality preference the lower and upper formulas meet at the same endpoint with their respective subgradient intervals. No auxiliary departure variable is needed to evaluate this row operation.

The derivation is a standard convex proximal calculation, consistent with the framework of Moreau and subsequent proximal methods [23, 27]. It does not establish that unequal side prices were previously inexpressible. DAQP's existing side-weight interface and an explicit finite-side lift already represent this objective [1]. The proposed computational property is the placement of the admitted update in the original-row native path and its potential effect on complete-operation cost. A claim of global priority would require stronger historical evidence than the construction or its numerical agreement supplies.

An equivalent residual description clarifies sign conventions. Let $C = [\ell, u]$, $r = z - \Pi_C(z)$ and

$$\Psi(r) = a_U(r)_+ + a_L(-r)_+ + \tfrac{1}{2}b_U(r)_+^2 + \tfrac{1}{2}b_L(-r)_+^2. \tag{6}$$

Then $\phi(z) = \Psi(z - \Pi_C(z))$. A corrective slack s used in $z + s \in C$ has the opposite sign and cost $\Psi(-s)$. Forgetting that reflection exchanges lower and upper prices while preserving an apparently convex objective. Such a solver can converge successfully to the wrong policy. The independent evaluator therefore reconstructs departure from the original row value and evaluates the original coefficients directly.

Dual consistency supplies another independent check. For finite endpoints and positive quadratic coefficients, the conjugate is

$$\phi^*(y) = \begin{cases} uy + [(y - a_U)_+]^2/(2b_U), & y \ge 0, \\ \ell y + [(-y - a_L)_+]^2/(2b_L), & y < 0. \end{cases} \tag{7}$$

When the corresponding b is zero, its quadratic conjugate term is replaced by the appropriate price-cap indicator. An absent endpoint restricts the corresponding multiplier sign rather than introducing infinite-times-zero arithmetic. These cases follow by maximizing the linear dual pairing over each exterior ray [30]. The relationship tests price scaling and sign recovery independently of the primal proximal formula.

For the full problem, original stationarity and subgradient consistency require

$$Px + q + A^\top y = 0, \qquad Ax = \operatorname{prox}_{g_{\text{all}}}^{H_0}(Ax + H_0^{-1}y), \tag{8}$$

where H_0 is an explicitly selected positive diagonal diagnostic metric with cost per squared row-unit entries, not the iteration's numerical penalty. A numeric identity is used only in canonical diagnostic coordinates. This diagnostic remains well defined at kinks. A small stationarity

residual alone is insufficient: an unconstrained stationary decision outside a priced interval can have a wrong row multiplier. Conversely, a nonunique dual need not match an oracle componentwise when it satisfies the correct original conditions. The comparison therefore retains objective, hard feasibility and method- appropriate dual diagnostics as distinct quantities.

The native split retains the original variables and row system. With positive primal regularization S and positive diagonal row metric H, the graph-proximal calculation involves

$$P + S + A^\top H A \succ 0. \tag{9}$$

Changing side prices changes the proximal map, while changing a numerical metric can change preparation or factorization. Unchanged dimensions do not imply that every update is free. The tested prepared workspace admits only fixed matrices, coefficient structures, hard mask and finite-side patterns, with the permitted vector updates. It refuses a changed model structure rather than reusing a factor that belongs to another problem.

Algorithm 1 states the executable procedure. It separates semantic admission, numerical iteration and original recovery. The separate engine owns the native update and its binding; the product does not emulate a solver in the browser. The original registered computation uses its exact recorded source rather than silently replacing it with a later public wheel. That source distinction is necessary for reproducible timing even when both installations implement the same mathematical penalty family.

Algorithm 1 Execute side-specific interval updates on the original row system, preserve hard policy, and return original-space quantities for independent objective and feasibility evaluation.

Require: Original convex QP, nonnegative side prices, positive metric
1: Refuse unsupported grouped/Huber prices and invalid row metadata
2: Prepare the original-row native linear system
3: **while** the numerical termination criterion is not reached **do**
4: Apply interval projection on hard rows
5: Apply (5) on admitted soft rows
6: Continue the native primal/dual splitting calculation
7: **end while**
8: Recover x, y and original lower/upper departures
9: Recompute objective, hard error and available dual diagnostics

Physical covariance is an additional semantic restriction. Under a positive row conversion $z' = dz$, equivalent prices and duals satisfy

$$a'_L = a_L/d, \quad a'_U = a_U/d, \quad b'_L = b_L/d^2, \quad b'_U = b_U/d^2, \quad y' = y/d. \tag{10}$$

The bounds and row matrix must also scale. Changing only the numerical coefficient or the unit label is not an equivalent problem. The concrete tests exercise original-space covariance and deliberately wrong units; they do not establish every floating-point quantity invariant under arbitrarily extreme scaling. Floating-point conditioning remains a separate numerical issue [17, 19].

The main implementation advantage sought was reduced setup and solve work against equivalent lifting. The original evaluation does not establish that advantage: its complete primary interval crosses zero. That outcome is consistent with a scalar operation being exact yet contributing little to total validated-call cost. Formulation construction, factorization, backend

maturity, input checks and recovery can dominate the saved auxiliaries. The implementation and its derivation remain useful for explaining side-specific semantics, but the failed prediction cannot be replaced with a favorable case, a selected dimension or a warm workspace example after the held-out results are known.

Finally, the convergence interpretation remains bounded. Fixed-parameter convex splitting under an attained saddle point supports convergence in its stated domain [11, 3]. The original solver may use documented adaptive numerical settings, whose observed successful outputs are tested directly. The scalar formula does not by itself prove every adaptive schedule, every numerical infeasibility certificate or a finite-precision error bound. Keeping that distinction explicit avoids turning a correct local minimization into a claim about all behavior of a large native optimization implementation.

The distinction between a finite linear price and exact hard feasibility can be stated more precisely. Suppose the corresponding hard-constrained convex problem has an attained primal and dual optimum. A linear price that dominates an appropriate optimal multiplier can support an exact penalty representation of its optimum [18]. Equality at the threshold can still allow additional elastic minimizers, so a finite user-selected coefficient cannot be interpreted as an unconditional physical guarantee. The product instead treats the hard mask as an independent invariant. This is why changing a cost factor never changes which original rows are immutable.

Proximal covariance also requires transforming the numerical metric when comparing individual row operations. Under the positive row conversion in(10), the temporary value and bounds scale with the row, and the equivalent metric scales inversely with its squared unit factor. Holding that metric fixed instead can change the numerical trajectory while preserving the original optimization target after full model conversion. Therefore equality of a single proximal iterate and equality of the final physical optimum are different claims. The tests state which object is being compared rather than using a relabelled axis as evidence of dimensional invariance.

Sparse validation similarly has several layers. A compressed-column object must have consistent pointers, row indices, shape and vector lengths. That structural admission does not prove that arbitrary supplied curvature is positive semidefinite. The mathematical method assumes a convex model, generated comparison cases have the documented curvature, and backend/independent numerical failures remain observable. Adding a small permanent diagonal shift to make another model easier would change its objective. The original profile therefore records numerical settings and scientific coefficients separately instead of treating both as interchangeable solver options.

The implementation comparison also separates timed and independent lifting. The timed lifted path uses the same pinned native backend family as the candidate, reducing one obvious implementation mismatch. The accuracy oracle uses stock OSQP with its own tighter settings. Substituting that oracle's measured time after seeing a result would alter the comparator. Likewise, comparing an already prepared candidate to a reference reconstructed on every repetition would answer a reuse question different from the registered cold-call hypothesis. All those boundaries remain explicit in the stored observation fields.

The scalar update's arithmetic is only one part of the full operation. Original objective evaluation requires recovering row values and pricing their actual departures. Imported or exported results must also preserve the base model, side coefficients and hard policy. A smaller sparse system may coexist with similar validation cost or additional conversion work. The numerical experiment measures the declared whole call; it does not time each arithmetic primitive independently. A proposed explanation that thresholding, validation or sparse ordering dominates

would consequently require additional instrumentation before becoming a measured causal conclusion.

Finally, the original primary sample has a specific dependence structure. Seven repetitions on each of twenty-four problems generate the reported paired observations. Several problems share one evaluation seed, which is why the confidence procedure resamples seed clusters. Treating every repeated time as a new independent problem would exaggerate information and answer a different statistical question. The native implementation's mathematical admission and the primary experimental denominator are both preserved, even though they constrain different parts of the resulting claim.

5 Proposed method: directional group-peak updates

The second construction couples interval departures within disjoint groups. A shared peak price can allocate corrections differently from independent row prices: changing the largest weighted departure alters the group cost, and several rows may become active at the same peak. Unequal upper and lower scales specify directional policy. The interface retains original interval rows while using an exact group proximal operation. Directional expressibility is not new: a one-sided row split or a compact epigraph already expresses the same objective. The research question concerns the resulting complete numerical operation.

For a nonempty closed box C and a residual penalty Ψ, define

$$g_C(z) = \inf_{s \in C} \Psi(z - s). \tag{11}$$

Assume Ψ is proper, closed, convex, nonnegative and satisfies $\Psi(0) = 0$. The relevant contraction property is

$$\Psi(Dw) \le \Psi(w), \qquad D = \operatorname{diag}(d_i), \quad 0 \le d_i \le 1. \tag{12}$$

This property permits unequal behavior on positive and negative rays. It does not require global sign symmetry. Orthant-monotonicity provides important prior context for that distinction [6]; convexity alone is not enough to justify the box projection identity below.

Theorem 5.1 (Box-residual identity)**.** *The contraction property*(12) *holds if and only if, for every nonempty closed box and every z,*

$$g_C(z) = \Psi(z - \Pi_C(z)). \tag{13}$$

Proof. For any $s \in C$, set $r = z - \Pi_C(z)$. If z_i exceeds its upper bound, then $z_i - s_i \ge r_i > 0$. If it is below its lower bound, then $z_i - s_i \le r_i < 0$. Inside the interval $r_i = 0$. Therefore $r = D(z - s)$ for some diagonal contraction. The property gives $\Psi(r) \le \Psi(z - s)$, with equality attained by $s = \Pi_C(z)$. Conversely, fix w, D and choose coordinate intervals with endpoints $-w_i$ and $-d_i w_i$ in increasing order. At $z = 0$ the nearest-box residual is Dw, while $-w$ is an admissible point. The claimed identity therefore implies $\Psi(Dw) \le \Psi(w)$. □

The theorem states an equivalence over all boxes. It is not merely a numerical observation on one training case. The infimal-convolution representation gives convexity. Its explicit composition with the continuous residual map supplies lower semicontinuity under the stated closedness assumption. A point inside the box has zero price, so the eliminated penalty is proper. These domain statements matter when an extended-real penalty represents an immutable restriction rather than a finite departure price.

Let $H = \operatorname{diag}(h_i) \succ 0$ be fixed. Define

$$\operatorname{prox}_f^H(v) = \operatorname{argmin}_z\{f(z) + \tfrac{1}{2}(z-v)^\top H(z-v)\}. \tag{14}$$

The translated residual relation is

$$\operatorname{prox}_{g_C}^H(v) = \Pi_C(v) + \operatorname{prox}_\Psi^H(v - \Pi_C(v)). \tag{15}$$

Proof. Put $p = \Pi_C(v)$ and clip a candidate coordinate to the segment joining p_i and v_i. Diagonal quadratic distance from v cannot increase under that clipping. The new box residual is a coordinate contraction of the original residual, so its penalty cannot increase either. An optimum can therefore be sought on that product of segments. Writing $z = p + w$ makes its residual exactly w. The unconstrained residual proximal minimizer lies on the same segments because clipping toward the segment from zero to $v - p$ again decreases both terms. Thus the segment restriction does not change the residual minimum, establishing the relation. □

Diagonal structure is essential to this proof. A dense metric couples coordinates, and coordinate clipping can then increase the quadratic distance. The theory section gives a concrete counterexample. Likewise, the identity is not claimed for an arbitrary convex price lacking the contraction property. The admitted directional group penalties satisfy it because reducing a coordinate magnitude without changing its sign cannot increase any scaled directional peak. Separate disjoint groups then permit independent group calculations within the diagonal metric.

For one group, define

$$M(w) = \max_i\{c_{U,i}(w_i)_+, c_{L,i}(-w_i)_+\}, \qquad \Psi(w) = aM(w) + \tfrac{1}{2}bM(w)^2, \tag{16}$$

with positive directional scales and $a, b \geq 0$. For residual $r = v - p$, choose c_i according to the sign of r_i. Fixing a candidate peak $t \geq 0$ constrains each residual coordinate to its corresponding interval. The best coordinate and scalar objective are

$$w_i(t) = \operatorname{sign}(r_i)\min\{|r_i|, t/c_i\}, \qquad F(t) = at + \tfrac{1}{2}bt^2 + \tfrac{1}{2}\sum_i h_i(|r_i| - t/c_i)_+^2. \tag{17}$$

The remaining problem is convex in one scalar. Breakpoints occur at $c_i|r_i|$. Sorting them in descending order gives active prefixes. On an interval with active set $S(t) = \{i : c_i|r_i| > t\}$, differentiating yields

$$t = \frac{\sum_{i\in S(t)} h_i|r_i|/c_i - a}{b + \sum_{i\in S(t)} h_i/c_i^2}. \tag{18}$$

Only a root consistent with its active interval is accepted. The zero endpoint is checked explicitly. When both prices vanish the original residual is returned directly; an empty active set never triggers a division by zero. Repeated breakpoints and zero residuals require consistent endpoint handling rather than arbitrary active-set choices.

The operation costs $O(k \log k)$ for sorting a group of size k, then a prefix scan. Weighted threshold and projection methods already employ such structures [28, 10, 7]. The implementation's specific role is to preserve the directional interval meaning, diagonal metric and original-row split, including boundary and representability cases. No claim identifies sorting or the scalar threshold primitive itself as an original algorithmic discovery.

Algorithm 2 makes those cases explicit. Numerical threshold comparisons use finite local spacing rather than a fixed absolute departure floor that changes meaning under unit conversion.

Nonrepresentable arithmetic is refused. The interface requires disjoint groups; treating overlaps as separate independent calls would in general change the objective's proximal operation. Positive finite scales are required, and zero group price is distinguished from a hard constraint. These are semantic conditions of the implemented method rather than details to infer from a successful backend return code.

Algorithm 2 Evaluate the directional disjoint-group proximal map with a fixed diagonal metric, retaining exact endpoint, zero-price and original-row semantics for independent numerical acceptance.

Require: Nonempty box, disjoint groups, positive scales and diagonal $H \succ 0$

Set $p = \Pi_C(v)$ and $r = v - p$
for each group **do**
  if both group prices are zero **then**
    Keep that residual group unchanged
  else
    Select each scale from the residual sign; sort $c_i|r_i|$
    Check $t = 0$ and consistent roots of (18)
    Recover w using (17)
  end if
end for
Return $p + w$, retaining exact hard-row projection separately

The joint conjugate is needed for independent acceptance. At fixed peak, maximizing the dual pairing produces the directional quantity

$$L(y) = \sum_i \left((y_i)_+ / c_{U,i} + (-y_i)_+ / c_{L,i} \right). \tag{19}$$

Taking the scalar conjugate gives

$$\Psi^*(y) = \begin{cases} (L(y) - a)_+^2 / (2b), & b > 0, \\ 0, & b = 0,\ L(y) \le a, \\ +\infty, & b = 0,\ L(y) > a. \end{cases} \tag{20}$$

The interval support term remains necessary: $g_C^* = \sigma_C + \Psi^*$. A single coordinate-wise price cap is therefore not a valid replacement for the group budget. The evaluator checks each group's absolute normalized Fenchel relation, raw and bounded-repaired stationarity, and every hard row's complementarity separately. A negative approximate hard-row gap cannot cancel an incorrect soft relation.

Finally, exact competing formulations delimit the computational claim. Directional one-sided rows reproduce the same peak under the published grouped projection path, while one nonnegative epigraph variable per group represents the convex price in an ordinary QP. The registered comparison retains both alternatives and direct/proximal DAQP policies where compatible. The primary operation includes 32 updates and all required recovery/acceptance work, not only one scalar threshold call. Its negative complete-population score demonstrates that a smaller original-row system can coexist with worse measured complete-operation cost. It does not refute the proximal identity, imply that every instance is slower by the same factor, or establish an advantage on an untested application regime.

Absent interval sides require explicit domain treatment. A missing upper bound removes its positive directional departure, and a missing lower bound removes its negative departure. It does not contribute an artificial large finite endpoint or a product involving infinity and zero. The interval support in the conjugate then restricts the corresponding dual sign. This links a modeling convention at the serialized input boundary to a mathematical condition in original-space acceptance. A primal answer with a plausible objective can still fail if its recovered multiplier assigns force to a side that does not exist.

Free group prices provide another useful boundary case. If both the linear and quadratic group coefficients vanish, the residual penalty is identically zero and the unconstrained residual proximal map is the identity. This is different from a hard box indicator, whose row operation projects to the box. The conjugate of the free residual penalty permits only zero soft-group dual contribution. These relationships give mutually consistent primal, proximal and dual controls. They prevent an endpoint special case from accidentally converting a free preference into an immutable constraint.

Within a nonzero-price group, the scalar threshold can make several coordinates meet the same weighted peak. A coordinate below the active threshold keeps its residual value, whereas a larger weighted residual is clipped to its directional peak allowance. This is a coordinated allocation, not an independent softening of each original row. At tied breakpoints several algebraic active prefixes can describe the same unique proximal vector. The implementation must recover that vector consistently rather than infer scientific nonuniqueness from an internal sorting tie. Empty groups and overlapping memberships are rejected by the actual interface.

The weights in(18) have physical meaning. The metric contribution divides by the squared directional scale, while the linear numerator divides by that scale once. Omitting one factor changes both threshold location and dimensional covariance. The independently expressed epigraph gives a different numerical route to the same group optimum, making this a concrete falsification target. The physical teaching input compares actual native, row-split and epigraph outputs and retains the original hard coupling equality. It is deliberately small enough to inspect without treating agreement as additional evidence of timing superiority.

The original-row formulation also does not imply constant preparation cost under every update. A change in a vector is different from a change in matrix structure, metric, group membership or penalty family. Only the registered repeated-forecast vectors are updated within a prepared operation. Hard dynamics and heater-energy limits remain immutable in the thermal family. A different group layout or unit mapping would identify another input batch and invalidate reuse of the original source-bound observation. This is why batch digests, units and method policy settings accompany each recorded operation.

Compatibility of references is similarly derived from semantics, not from an optimistic output flag. The linear row-split reference cannot be used as if it handled every quadratic group price. A direct DAQP epigraph requires its admitted curvature conditions; documented proximal policies have separately fixed roles. The complete protocol retains excluded cells as incompatible and dependent oracle limits as oracle-failed. Consequently the fastest selected reference is defined by the registered compatible alternatives for that instance, and cannot be chosen from an arbitrary easier objective afterward.

All those conditions concern the same optimization model from different directions: input units, convex proximal structure, equivalent formulations, original duals and complete-operation accounting. Their agreement establishes a restricted and reproducible implementation claim. It does not authorize transferring a favorable teaching example to the entire held-out timing

population. The retained negative score includes all required members, including those whose candidate or reference operation did not yield an accepted complete answer. That distinction preserves the scientific question even when a local closed-form calculation is exact.

6 Theory and its limits

The local proximal identities must be separated from convergence of a full splitting algorithm and from finite-precision numerical acceptance. Each admitted scalar or directional proximal problem has a proper closed convex objective plus a positive quadratic metric. It therefore has a unique minimizer. The original QP itself may still be infeasible, unbounded or nonunique. A local row formula cannot remove those global possibilities. Standard proximal and monotone-operator results provide the appropriate framework for the distinction [27, 31, 21, 11].

For the grouped split define a graph-constrained function

$$F(x,z) = \tfrac{1}{2}x^\top Px + q^\top x + \iota_{\{z=Ax\}}(x,z), \tag{21}$$

and set $G(x,z) = g_{\mathrm{all}}(z)$, which contains the hard intervals and the admitted soft costs. The fixed block metric is $\mathcal{M} = \mathrm{diag}(S,H)$ with $S \succ 0$ and positive diagonal H. The graph subproblem is uniquely defined because its reduced matrix $P + S + A^\top HA$ is positive definite. This is numerical regularization of a proximal subproblem, not an undisclosed addition to the scientific objective. The target problem remains the original constrained convex optimization model.

Write J_F, J_G for the proximal resolvents in this fixed metric and $R_F = 2J_F - I$, $R_G = 2J_G - I$ for their reflectors. The native consensus state uses the scaled dual d and the following four updates:

$$\begin{aligned} u^{k+1} &= J_F(v^k - d^k), \\ h^{k+1} &= \alpha u^{k+1} + (1-\alpha)v^k, \\ v^{k+1} &= J_G(h^{k+1} + d^k), \\ d^{k+1} &= d^k + h^{k+1} - v^{k+1}. \end{aligned} \tag{22}$$

Define $s^k = v^k + d^k$ after a complete step, so $v^k = J_G(s^k)$. Substitution with this specific state gives

$$s^{k+1} = \left(1 - \frac{\alpha}{2}\right) s^k + \frac{\alpha}{2} R_F R_G(s^k), \qquad 0 < \alpha < 2. \tag{23}$$

Under an attained saddle point and the stated closed-convex assumptions, the standard fixed-metric splitting theory supplies convergence of the appropriate shadow variables. The relaxation range makes the displayed operator averaged in the relevant metric. Its fixed points are connected to primal/dual solutions through the resolvents, rather than being identified with raw numerical coordinates without that recovery. This interpretation follows established splitting results [3, 11].

In original row variables, writing S for the positive primal metric and $y = Hd_z$, the same graph solve and relaxed updates take the form below. Since G is independent of x, its proximal update in that coordinate is the identity, and the scaled multiplier d_x is zero after a complete

step. The native initialization sets it to zero.

$$\begin{aligned}(P+S+A^\top HA)\widetilde{x}^{k+1} &= Sx^k - q + A^\top(Hz^k - y^k),\\ x^{k+1} &= \alpha\widetilde{x}^{k+1} + (1-\alpha)x^k,\\ z^{k+1} &= \operatorname{prox}^H_{g_{\mathrm{all}}}\left(\alpha A\widetilde{x}^{k+1} + (1-\alpha)z^k + H^{-1}y^k\right),\\ y^{k+1} &= y^k + H\left(\alpha A\widetilde{x}^{k+1} + (1-\alpha)z^k - z^{k+1}\right).\end{aligned} \tag{24}$$

The price function gains no relaxation factor. The grouped native implementation fixes a finite relaxation in the stated open interval and rejects adaptive penalty updates, automatic OSQP scaling, polishing, directional sensitivities and directional code generation. Its explicit row normalization is separate: $\bar{A} = DA$, $\bar{\ell} = D\ell$, $\bar{u} = Du$, $\bar{c} = c/D$, $y = D\bar{y}$ and $H_{\mathrm{physical}} = D\bar{H}D$. Setup chooses $D_i = \max(c_{L,i}, c_{U,i})$ on soft rows and one on hard rows. This identifies the physical metric; it does not claim that a previously unnormalized iteration has the same trajectory.

It is not a rate theorem for every admitted semidefinite QP. Stronger contraction or linear-rate conclusions need additional quantitative assumptions [14]. Adaptive changes in the row metric or the relaxation parameter produce a different iteration and require their own conditions. The empirical original protocol's documented adaptation is evaluated numerically; the fixed-metric argument is not presented as a proof of every adaptive schedule in every implementation. Likewise, iteration-limited candidate vectors remain subject to the external acceptance rule regardless of their distance from a recent iterate.

The diagonal assumption in the translated proximal identity cannot be discarded. Consider $C = (-\infty, 0] \times \mathbb{R}$, $\Psi = \iota_{\{0\}}$, $v = (1, 0)$ and

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. \tag{25}$$

The residual formula would return the Euclidean box projection $(0, 0)$. However, the true H-metric projection is $(0, 1/2)$: fixing the active first coordinate at zero, differentiation of the quadratic in the second coordinate gives $-1 + 2z_2 = 0$. The metric is positive definite, and the residual indicator satisfies coordinate contraction monotonicity, yet the identity fails. Positive definiteness alone is therefore not an adequate replacement for a positive diagonal metric in that theorem.

Equivalent formulations provide additional mathematical controls. A directional upper row is $c_{U,i}A_ix \le c_{U,i}u_i$, while a lower row is $-c_{L,i}A_ix \le -c_{L,i}\ell_i$. Their positive departures produce the same directional peak. A compact epigraph introduces $t_g \ge 0$ and bounds every scaled departure by t_g, charging $a_g t_g + \frac{1}{2}b_g t_g^2$. Nonnegative prices permit a tight epigraph at an optimum, with harmless nonuniqueness when the price is free. These are exact objective representations, not weakened reference problems. They motivate independent numerical comparisons without providing a theorem about which representation is fastest.

Finally, original-space acceptance includes assumptions not proved by an algebraic identity alone. Sparse source conversion, finite precision, dual recovery and physical units can be wrong even when the intended formula is correct. Independent lifting, local Fenchel relations, hard-row complementarity and reciprocal scaling controls test those implementation boundaries. Such checks supply finite evidence on actual inputs. They do not replace the mathematical domain restrictions, nor establish operational safety, global priority or a complete-operation timing gain.

7 Experimental design

Two prospective protocols test two distinct computational predictions. The original separable study predicts at least 10% median relative reduction in cold validated-call time against the faster admitted lift or DAQP reference in its specified asymmetric-elastic regime. The grouped study predicts a corresponding improvement in a complete repeated-forecast operation against its strongest compatible registered reference. Each prediction also requires a positive lower confidence limit and complete accuracy-qualified evidence. Neither protocol permits choosing a favorable subgroup after observing timings.

The original study has 456 seeded instances: a main grid of dimensions 32,64,128 and 256, densities 0.05 and 0.2, diagonal conditions 1 and 1000, and symmetric-elastic, asymmetric-elastic and asymmetric-linear families, plus a separately specified robustness grid. Its six evaluation seeds differ from development seeds. The primary regime contains 24 distinct held-out asymmetric-elastic problems with dimension at least 128 and density 0.05. Seven measured repetitions produce 168 paired observations, not 168 independent generated problems. Each backend has one excluded warm-up, and method order is randomized by a fixed seed within each instance and repetition.

Original cold timing includes validation, formulation, setup, solve and original-space evaluation while excluding generation, module import and export. Solver tolerances are10^{-7}; the stock-lift oracle uses10^{-9}. External objective discrepancy and raw hard error must not exceed10^{-6}. Objective discrepancy is normalized by the larger of one and the oracle objective magnitude. Stationarity is diagnostic in this protocol, not a newly imposed acceptance threshold. The exact native source and fixed numerical settings are retained, including the distinction between its historical source profile and later public wheel installations.

For observation j, with native time $T_{N,j}$ and selected compatible baseline time $T_{B,j}$, the paired quantity is

$$R_j = 1 - T_{N,j}/T_{B,j}. \tag{26}$$

The original endpoint takes the median over its registered repeated paired observations. For the grouped endpoint, take each policy's median over the three measured operations on a given instance. Set T_B to the minimum among the compatible registered baseline medians and score the instance as $1 - T_N/T_B$. If any required compatible measurement is missing, unsuccessful, timed out or rejected for accuracy, assign the entire instance a score of -1. Prespecified incompatible policies contribute no measurement. Qualification requires a median of at least 0.10, a positive lower confidence limit and zero failed pairs.

The two endpoints resample six complete seed clusters with replacement. The original analysis uses 2,000 resamples and seed 314159. The grouped analysis uses 10,000 resamples and seed 701119, pooling the 24 instances from each drawn seed, taking the median and using the 2.5th and 97.5th percentiles with linear interpolation. Dependence among repetitions makes cluster identity important, while few-cluster coverage remains a limitation [12, 8, 5]. Table 2 identifies the generator and order seeds.

The grouped study contains 48 development,144 held-out and 18 robustness inputs across sparse and synthetic hourly thermal families. The complete held-out 144-member population is the primary denominator. Nine method policies use one warm-up and three measured operations per input; an oracle observation is also retained. Thus 210 inputs produce 7770 raw observations and 5670 measured method observations. Each inner timed operation includes preparation/factorization, all 32 vector updates and solves, physical recovery and acceptance cal-

Table 2: Generator seed blocks for each population and the method-order randomization seeds for the two fixed protocols. Ranges are inclusive.

Role	Original	Grouped
Development	1000–1002	710001–710002
Held out	9000–9005	910101–910106
Robustness	8500–8505	810101–810103
Method order	271828	601117

culations. Compiled competitors are symmetrically preloaded before that timer. Process/library startup and serialization are excluded from inner timing but retained by the outer resource supervisor.

The grouped references include stock-OSQP compact epigraphs, directional row-split generalized projection, direct DAQP epigraphs and documented proximal DAQP variants [2]. Compatibility depends on the original curvature and price family. Every candidate batch requires all 32 frames to return and pass. In physical units the maximum hard-row departure must be at most 10^{-6}. The objective discrepancy uses $\max(1, |F_{\text{oracle}}|)$ as its denominator. For each of the raw and repaired multipliers v, the stationarity denominator is $\max(1, \|Px\|_\infty, \|q\|_\infty, \|A^\top v\|_\infty)$. For a soft group, the Fenchel defect is divided by the maximum of one and the magnitudes of the group penalty, box support and group dual pairing. For a hard row, the corresponding maximum uses one and the magnitudes of the signed multiplier times its active bound and times the returned row value. The objective, both stationarity checks, each local normalized Fenchel defect and the joint dual-budget check must be at most 10^{-6}. For $b_g = 0$, linear price a_g and raw polar value p_g, the budget excess is $(p_g - a_g)_+ / \max(1, p_g, a_g)$. Dual repair, measured by $\|y - \bar{y}\|_\infty / \max(1, \|y\|_\infty)$ for repaired multiplier $\bar{y}$, is bounded by 10^{-7}. Signed cancellation between defects does not certify a frame. The common compact-epigraph oracle uses stock OSQP with $\epsilon_{\text{abs}} = \epsilon_{\text{rel}} = 10^{-10}$ and at most 200,000 iterations. Its independent original-space checks and repair use 10^{-7} limits. The recorded states distinguish incompatible cells, oracle limits, iteration-limited candidates and timeouts.

Retained component comparisons cover formulation choice, numerical conditioning and common-trajectory cold/warm reuse. They were not randomized interventions that independently disable each implementation feature. The ablation section therefore reports their actual scope and explicitly identifies unperformed interventions. Standard optimization benchmark guidance motivates this separation of accuracy, success and runtime [9, 16]. The scientific inputs remain synthetic and the host nonreserved; neither prospective registration nor repeated execution turns the timing findings into an operational control or field-energy evaluation.

8 Results

Table 3 gives the two stored primary endpoints without pooling their populations. The original computation completed all 456 instances, retaining 11592 accepted and 7560 incompatible measured observations with no numerical or external-accuracy rejection. Its 168 paired observations yield 0.499839% median relative reduction, with 95% limits -3.291967% and 3.944835%. Figure 1 displays this interval and the registered ten-percent reference. The interval crosses zero, so the original prediction is unestablished. The result is not replaced by the best-performing dimension or a selected repetition.

Table 3: Distinct primary populations, stored effect estimates and uncertainty for the two completed timing studies. Failed-pair rules are not transferred between experiments and neither registered ten percent improvement criterion is established.

Study	Population	Pairs	Effect (%)	95% limits
Separable	456	168	0.499839	-3.29197, 3.94484
Grouped	210	144	-212.45	-225.337, -192.874

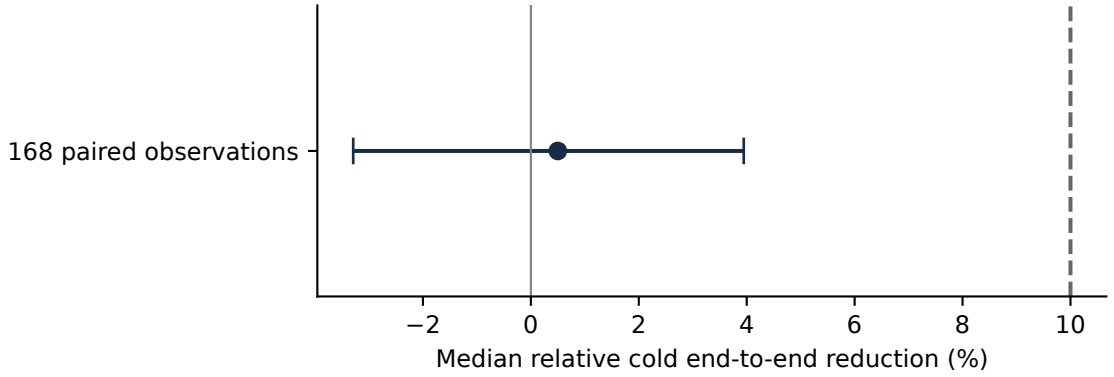


Figure 1: Original held-out timing estimate with its stored 95 percent seed-cluster confidence interval and registered ten percent reference. The interval crosses zero; neither interval nor population is recomputed for this figure.

Figure 2 relates the original scalar objective to the actual returned method optima. Agreement outside the preferred interval is consistent with the declared finite departure price. The analytic curve is an evaluation of stored coefficients, not another timed trial. Figure 3 instead displays stored cell medians and interquartile ranges in one asymmetric-elastic slice. It shows concrete complete-call costs while retaining its descriptive status. Formulation size and a selected timing curve cannot establish a general factorization or algorithmic advantage.

The original conditioning comparison in Figure 4 retains the maximum accepted objective discrepancy for each compatible robustness cell. The finite grid extends through diagonal condition10^6 without violating the registered objective/hard limits in admitted observations. This is not a proof for every near-dependent or arbitrarily scaled problem. The common-trajectory stream in Figure 5 shows stored cold and warm durations; it has one observation per frame and does not support an independent statistical superiority claim. Its component interpretation is examined in the next section.

The grouped study completed all 210 canonical inputs. Its 5670 measured method observations comprise 3667 accepted,12 accuracy-rejected, 1719 incompatible,270 dependent oracle-failed and two supervised timeouts. Table 4 and Figure 6 retain those categories separately in development, held-out and robustness splits. The 7770 raw observations additionally retain warm-ups and oracle calls. All 163390 returned candidate frames satisfy the applicable original- space checks;44 noncandidate backend-limit frames remain distinct. Framewise mathematical agreement does not override a failed operation's batch-level state or the primary failed-pair treatment.

The complete 144-member grouped endpoint is shown in Figure 7. Nine failed pairs retain their prespecified score; six missing native medians remain gaps. No reference is selected where

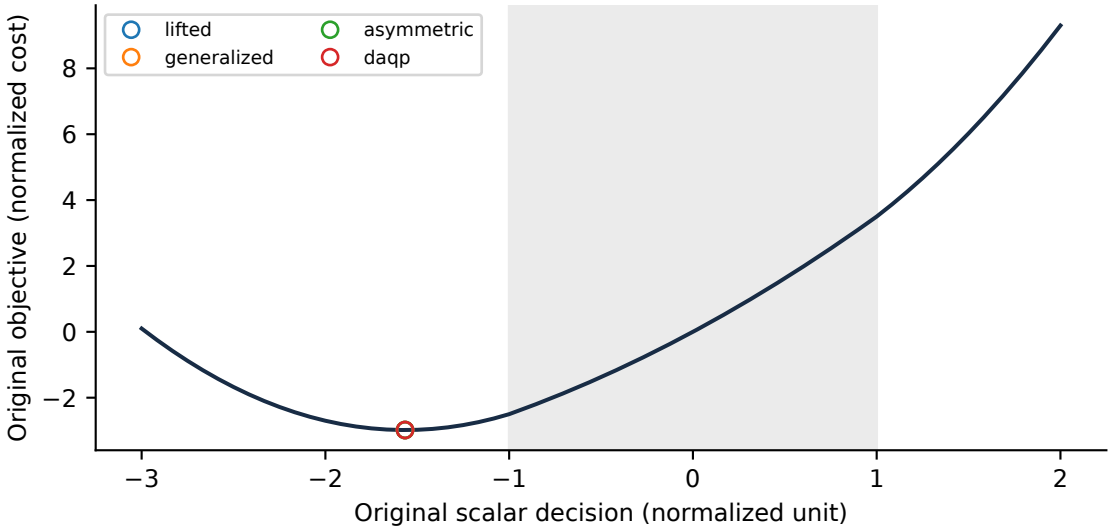


Figure 2: Analytic evaluation of the stored C01 objective and actual exported method optima. The shaded interval is the preferred band, and the curve is a mathematical illustration rather than a new timed observation.

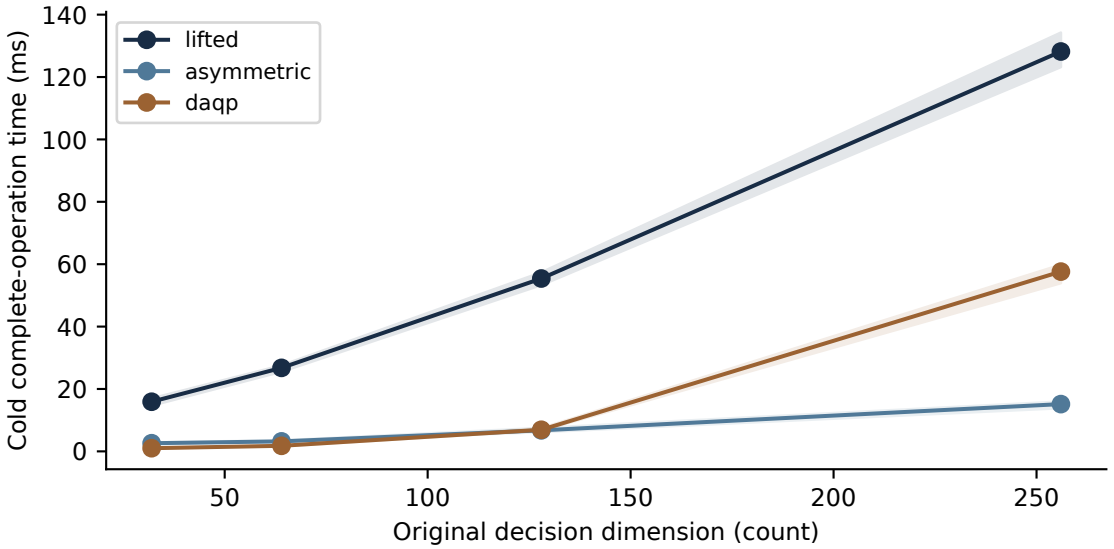


Figure 3: Stored held-out asymmetric-elastic timing cells at density 0.05 and unit diagonal condition. Lines are stored medians and bands stored interquartile ranges; this descriptive subset does not replace the complete primary endpoint.

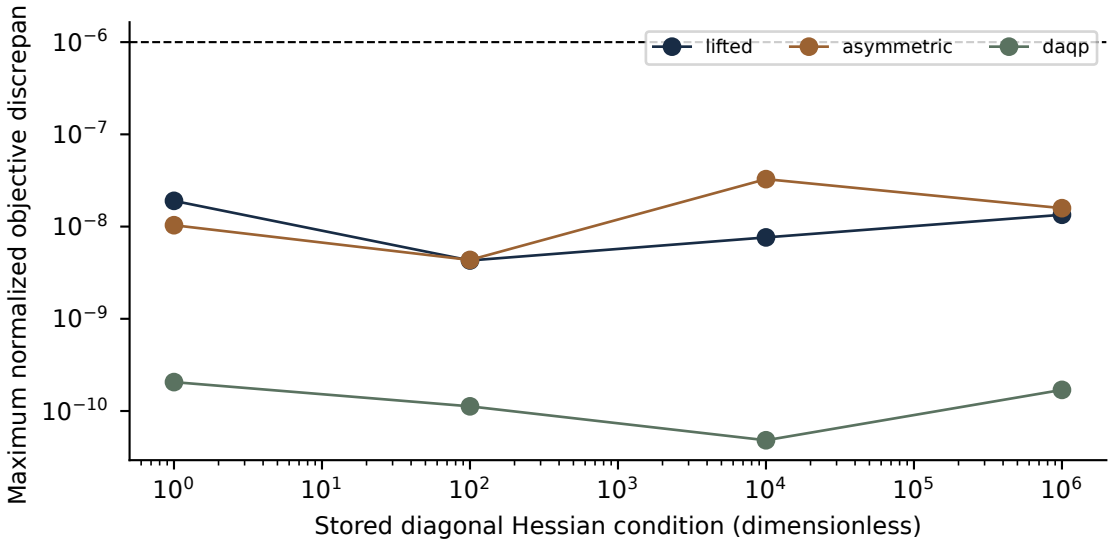


Figure 4: Maximum stored objective discrepancy in each compatible robustness cell by diagonal condition. The dashed line is the original external objective criterion; absent methods remain absent rather than receiving an invented zero discrepancy.

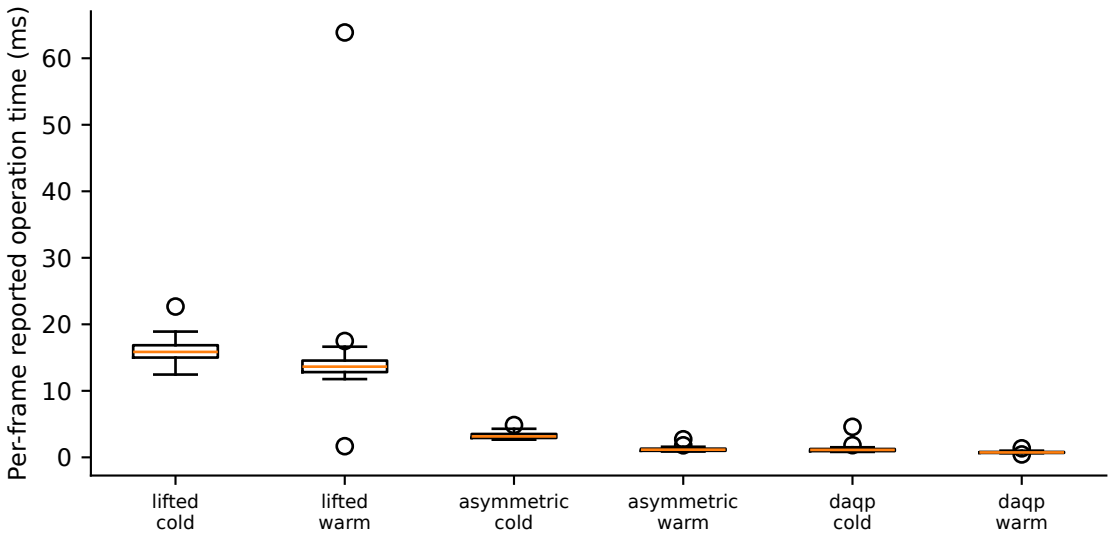


Figure 5: Stored cold and warm C12 operation durations for every accepted frame of each compatible method. The box summaries are descriptive paired-path evidence, not a randomized causal isolation of warm starting from all other implementation choices.

Table 4: Measured grouped-study method states by split, including dependent oracle failures and supervised timeouts. Counts cover all measured method rows, while warm-up and oracle observations remain in the larger raw archive.

Split	Accepted	Accuracy	Incompat.	Oracle	Timeout
development	879	0	396	21	0
heldout	2587	6	1188	105	2
robustness	201	6	135	144	0

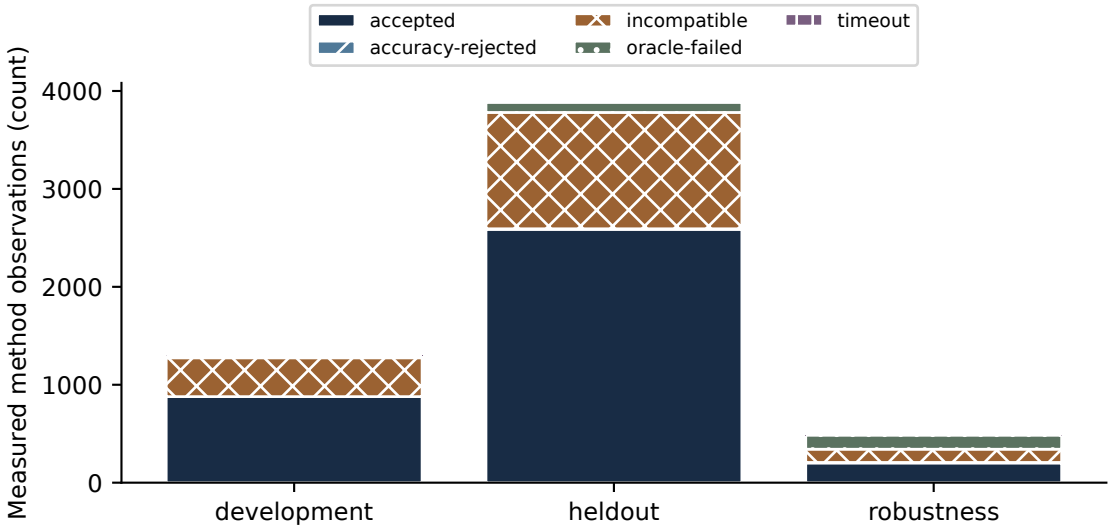


Figure 6: All 5670 measured method observations of the grouped study, separated by their original development, held-out and robustness splits. Acceptance, accuracy rejection, incompatibility, dependent oracle failure and supervised timeout remain different categories.

the primary rule selected none. The median paired score is -212.449875%, with 95% limits -225.336618% and -192.873759%. These percentages do not describe a uniform slowdown of every instance, and the result does not become favorable by discarding unsuccessful members. Figure 8 shows returned per-instance method medians with their differing sample counts. Those unmatched distributions characterize concrete formulations but are not an alternative primary estimator.

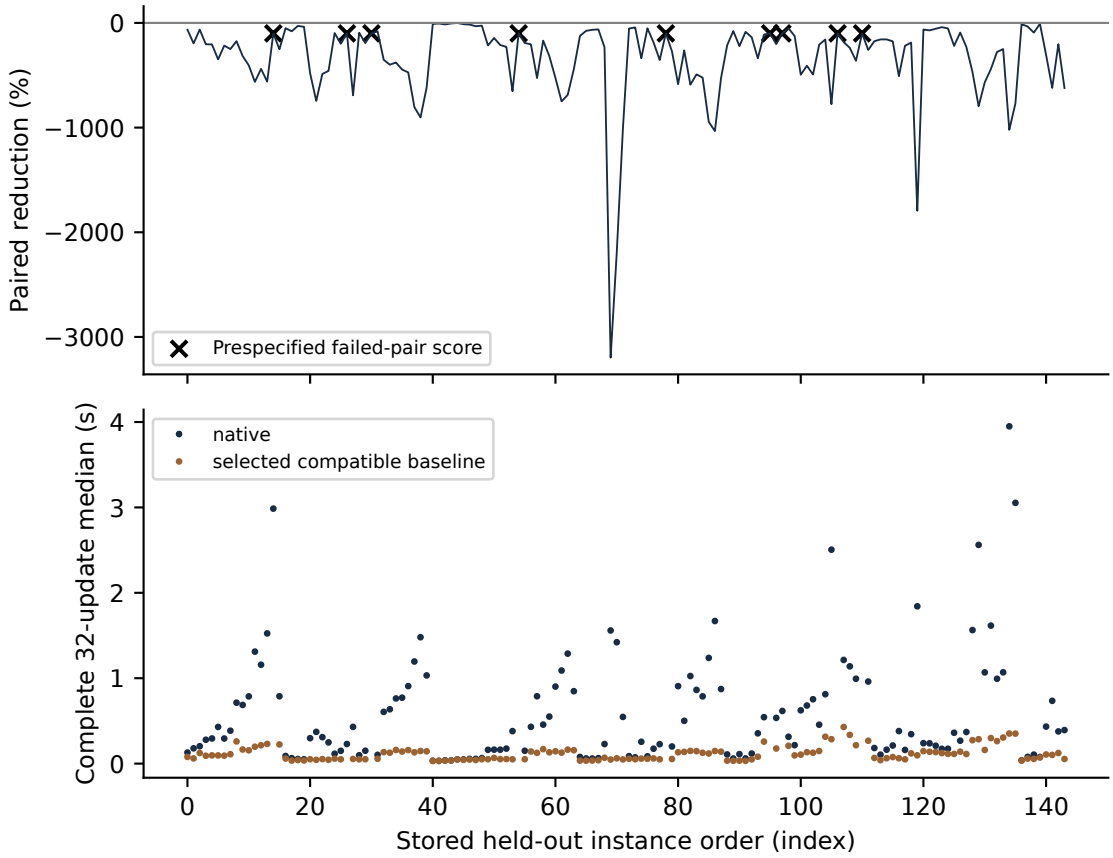


Figure 7: All 144 stored grouped-study primary members in original order, with nine failed scores retained. Complete-operation medians preserve missing values as gaps and do not impute a comparator where the primary rule selected none.

Figure 9 supplies a transparent original- space agreement example from the stored physical teaching input. Native, compact-epigraph and directional row-split answers agree on objective and decision within their numerical criteria. The example supports the intended formulation equivalence; it adds no timed observation to either registered population. Its role is to separate a mathematical or implementation correctness claim from the computational hypothesis that the full studies refute. Thus the results are consistent: exact proximal identities and small accepted examples coexist with negative complete-operation timing findings on the specified tested populations. The twelve accuracy-rejected batches include backend or iteration-limit and absent-vector outcomes, rather than twelve returned vectors failing the independent KKT calculation. Six held-out row-split batches returned status seven without a candidate vector. Native and forced-proximal DAQP operations account for the two timeouts. Six robustness

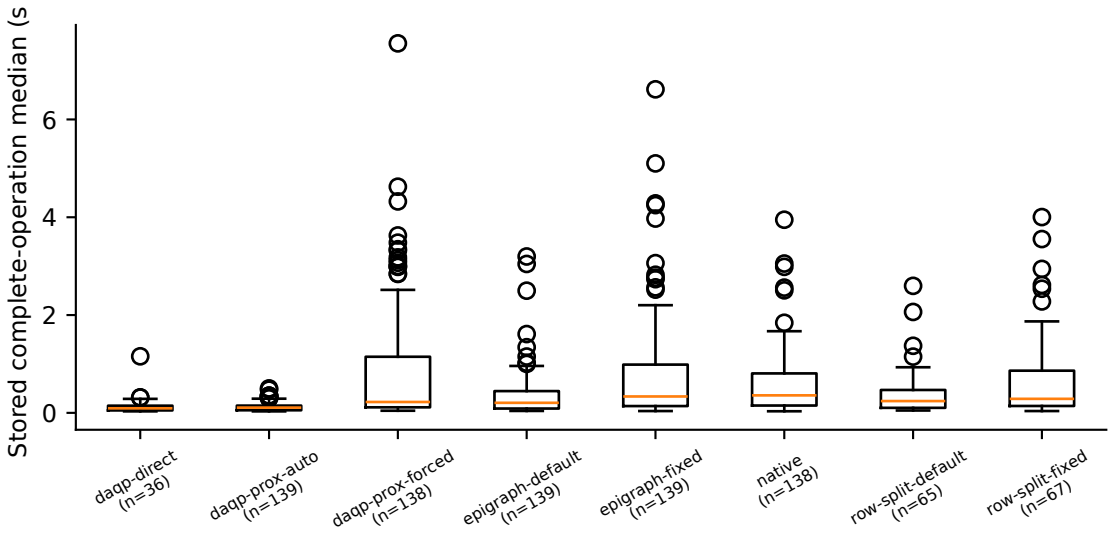


Figure 8: Distributions of stored held-out per-instance medians for each returned compatible formulation. Sample counts differ because compatibility and failures differ; these unmatched descriptive distributions are not the registered paired comparison or a causal ablation.

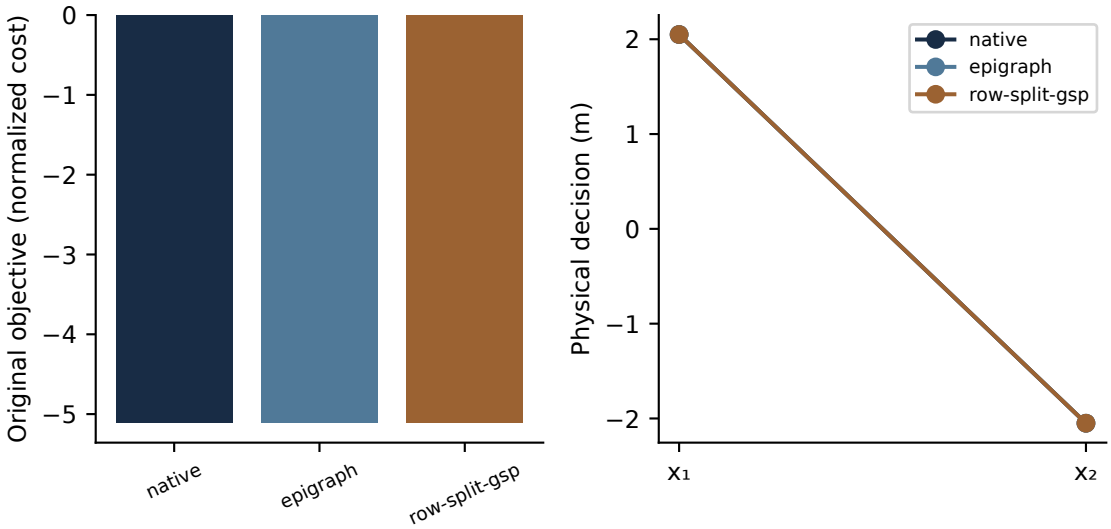


Figure 9: Stored native, epigraph and directional row-split answers for the same two-variable physical teaching input. Objective and decision agreement are a small numerical illustration, not additional timing evidence or an independent experiment.

oracle limits leave 144 dependent measured rows with oracle-failed state in that split. Those dependency counts are distinct from the nine failed primary pairs. All returned candidate frames passed their applicable original-space checks; a batch-level backend status remains a separate acceptance condition.

9 Ablation and retained component comparisons

A causal ablation would intervene on one implementation feature while preserving the same problem, inputs, acceptance criteria and other numerical choices. The retained evidence does not perform every such intervention. Table 5 instead states which component comparisons actually exist and what they can support. This distinction prevents a required analysis section from being filled with an invented experiment or an unsupported causal explanation for the primary timing outcome.

Table 5: Scope of retained component comparisons and unperformed causal interventions. Each row distinguishes recorded numerical evidence from a stronger attribution that the existing experimental design does not establish.

Component	Existing comparison	Inference limit
Representation	Native, compact epigraph and row-split policies	Complete policies differ; no isolated causal factorization effect
Workspace reuse	Common C12 cold/warm observations	No independent intervention on every initial iterate, cache and active set
Conditioning	Fixed-within-seed diagonal condition sweep	Finite robustness grid, not arbitrary physical scaling
Directional policy	Same-input equivalent teaching formulations	Correctness illustration, not timed performance or field benefit
Threshold implementation	Native exact prefix operation	No alternative sorting implementation timed in isolation
Acceptance overhead	Included original-space checks	No post-hoc removal to improve the registered endpoint

Table 6: Matched descriptive contrasts from retained observations. Positive timing reduction means the second configuration took less time. Conditioning rows report the change in normalized objective discrepancy.

Contrast	Matched pairs	Median paired effect
C12 cold to warm: Lifted OSQP	32	15.60%
C12 cold to warm: Original-row OSQP	32	65.09%
C12 cold to warm: DAQP	32	31.37%
Grouped epigraph to row split (default)	65	14.19%
Grouped epigraph to row split (fixed)	67	3.52%
Condition 1 to 10^6: Lifted OSQP	42	-1.60×10^{-9}
Condition 1 to 10^6: Original-row OSQP	42	5.95×10^{-11}
Condition 1 to 10^6: DAQP	42	-3.57×10^{-11}

Table 6 matches the retained records within the same input. The timing reduction is the median of paired $100(1 - T_{\text{second}}/T_{\text{first}})$ values, never a ratio of two marginal medians. For C12 the

pair is a frame; for grouped formulations it is a held-out instance with accepted medians for both policies. The conditioning diagnostic matches the original asymmetric-elastic robustness records at conditions 1 and 10^6 by seed and measured repetition. It reports the median within-pair change in normalized objective discrepancy. These descriptive subsets do not replace either complete registered endpoint.

The formulation comparison retains exact objectives. Row splitting changes row count and scaling while using a published grouped projection backend; a compact epigraph changes variable structure and can interact differently with sparse ordering and numerical preparation. The returned time distributions therefore compare complete reference policies. They do not isolate one matrix-size effect or prove that a threshold sort explains every difference. The unequal compatibility and failure counts in Figure 8 further prevent reading its unmatched boxes as a causal ranking on one common successful subset.

Cold/warm reuse provides a different component view. All methods receive the same oracle-driven C12 states, so differences are not caused by independently divergent plant trajectories. However, warm mode may retain factors, active sets and iterates together, while setup accounting also changes. The stream has one observation per frame and remains descriptive. It cannot assign the entire reduction to any one retained object or supply the primary confidence interval. Online active-set literature already uses parametric structure [13]; the observation is not evidence that reuse was introduced by this implementation.

The conditioning sweep changes diagonal curvature within its registered seed controls. It supports a bounded statement about objective/hard-error behavior on that grid. The dimensional teaching checks test a different invariant: reciprocal row and directional-scale conversion should preserve physical meaning. Changing a directional scale alone changes the objective's allocation policy. Neither comparison is a substitute for an unperformed randomized intervention on threshold arithmetic or for a new registered timing study. The fixed primary outcomes therefore remain the inferential conclusions, while these component analyses explain what future mechanism-specific experiments would need to hold constant.

10 Discussion

The central result is the coexistence of exact local mathematics and negative complete-operation timing evidence. The original-row updates preserve selected formulations without explicit departure variables in the principal system. That algebraic property is verified by independent equivalent formulations and original-space checks. The registered experiments nevertheless fail to establish their proposed timing gains. There is no contradiction: a smaller principal representation is only one contributor to preparation, iteration, recovery and validation cost.

The grouped operation adds a sorted threshold and joint dual checks. A sparse reference may benefit from simple diagonal auxiliaries or effective symbolic ordering. A prepared active-set backend may exploit vector updates efficiently. Dense/sparse conversion and language-level overhead can also matter. Without separate controlled instrumentation, these are plausible mechanisms to investigate rather than measured causal decompositions of the observed difference. Numerical optimization and floating-point analysis caution against inferring cost or accuracy from dimension alone [25, 17].

Including the strongest compatible formulations is therefore important. A symmetric grouped projection backend after directional row splitting can solve the same directional objective. An epigraph can represent the same group price with one auxiliary per group. The computational

comparison would be weaker if it omitted either because its representation differs from the candidate. Similarly, DAQP's existing side weights prevent an unequal-cost example from supporting a new expressibility claim. Prior orthant and weighted projection work constrains the interpretation of the analytic construction as well as its implementation [6, 28].

Operational interpretation requires another separation. A finite price can intentionally purchase an interval departure. Numerical optimality does not prove that the chosen price reflects a real engineering risk or that the resulting physical policy is safe. Predictive control relies on model, constraints and closed-loop assumptions beyond solving one convex subproblem [22, 29]. The source-faithful and authored building examples are synthetic, and the repeated forecasts are not measured building savings. Their value here is transparent interpretation and repeatable numerical comparison, not a field-performance claim.

The negative results constrain subsequent research. A revised method should identify a genuinely different computational mechanism, state why the strongest reference might fail to exploit it, and fix a prospective population before measurement. Selecting favorable current instances, weakening the oracle, removing acceptance work or changing the failed-pair rule would answer another question. The complete retained outcomes instead make it possible to formulate that next question without misrepresenting either study's original evidence.

Within the stated domain, the proximal formulas remain useful tools for modeling and independent verification. Their explicit zero-price, endpoint, sign and metric conditions clarify when an implementation is solving the intended problem. The present evidence supports those restricted mathematical and computational statements. It supplies no global priority claim, general runtime superiority or guarantee of practical benefit outside the tested synthetic regimes.

11 Limitations

Internal validity depends on exact objective matching and complete timing boundaries. The two protocols use different native profiles, repetitions, populations and operation scopes. Their primary percentages cannot be pooled or compared as though they measured the same unit of work. The original 168 paired observations include repeated measurements of 24 problems; the grouped primary contains 144 distinct held-out instances. Both confidence procedures have only six seed clusters. Cluster resampling preserves a selected dependence structure, but small-cluster uncertainty can remain substantial [5].

External validity is limited to the synthetic models and one nonreserved execution host. Fixed seeds and a supervised process do not eliminate operating-system interference, thermal behavior, library maturity or differences between Python/SciPy and native implementations. The tested sparse dimensions, curvature patterns, thermal horizons and directional groups do not sample every industrial QP. The models contain no measured customer data, field identification uncertainty or operational deployment evaluation. A negative result in this population is not proof that original-row updates cannot help another prospectively defined regime.

Construct validity requires keeping formulation, correctness and speed separate. Exact objective agreement does not establish an algorithmic advantage, while a purchased soft departure does not establish physical failure when that departure is the stated optimization policy. Conversely, a relaxed feasible representation does not certify satisfaction of original immutable inequalities. The reported objectives use model-specific normalized or physical cost conventions; they cannot be compared across unrelated cases as realized business benefit.

The theoretical domain is deliberately restricted. Box residual elimination requires contraction

behavior; the proximal identity uses a fixed positive diagonal metric. The native grouped interface has disjoint groups and positive finite scales. The convergence interpretation assumes a saddle point and admissible fixed relaxation. Dense metrics, overlapping groups, nonconvex prices and arbitrary adaptive schedules are outside the proved scope. Finite tests and local arithmetic safeguards do not constitute a universal finite-precision error theorem.

Finally, the component analysis retains existing comparisons instead of fabricating a full causal ablation. No independent sorting implementation or post-hoc removal of acceptance work was timed. Learned FlexQP parameters were unavailable, so the fixed reconstruction is not a learned-policy reproduction. These limits are part of the evidence interpretation and remain visible alongside the complete failed primary criteria.

12 Reproducibility

The public product source is https://github.com/fsantibanezleal/CAOS_SoftQP, with native code in the separate https://github.com/fsantibanezleal/CAOS_SoftQP_Engine repository. The original computation retains its exact historical source identity; the grouped computation retains its registered source, protocol and public SDK profile. These identities must not be replaced by a later wheel merely because the method name matches.

The complete grouped-study evidence is archived at https://doi.org/10.5281/zenodo.23875194. The product evidence tree is https://github.com/fsantibanezleal/CAOS_SoftQP/tree/2d3f299d80edad49c2599e61bb6e3c875c119043. Its `data/derived/experiments/campaign.json` and `data/derived/experiments/stream.json` contain the original exported summaries and common trajectory, while `manuscripts/coupled-peak-study/results/summary/summary.json` contains the grouped summary. The original native engine source is `d0c69ed974d577ba02adbd4806332270d9425651`; the grouped registered implementation source is `246bf87e690f4d010494c556e2021d311d587422`. The archive preserves the latter study's protocol and complete unsuccessful and successful observations, rather than only the displayed primary statistic.

The archived inputs include synthetic generators or canonical batches, seeds, full observations, unsuccessful states, method settings, source/environment identities and replay checks. The manuscript asset generator reads immutable stored objects and writes vector figures, literal tables and a source digest manifest. It performs no numerical solve, timing rerun or bootstrap recomputation. The separate native build command uses pdfLaTeX and BibTeX and refuses unresolved or clipped output. From the product repository root, install the figure dependencies, generate the evidence-derived assets and build the paper with these commands. Choose a fresh output directory outside the source checkout.

```
python -m pip install -r manuscripts/shared/requirements.txt
python manuscripts/shared/generate_assets.py
python manuscripts/coupled-peak-study/build.py --output /tmp/softqp-paper
```

The original recorded execution took 1363.063 s, with 92028928 bytes peak worker RSS and zero restarts. The archive does not identify the processor model. This process duration includes work beyond the registered per-call estimator. Source, mathematical and rendered-page review are distinct from successful compilation.

Replaying the preserved scientific evidence and rebuilding the figures does not require private operational data or API credentials. A fresh timing reproduction requires the exact registered

historical source/environment and a new external output location; it must not overwrite the original evidence. The physical teaching exchange is an additional transparent illustration and contributes no observations to either registered primary population.

Declaration of generative AI use. Generative AI tools assisted source synthesis, mathematical checking, implementation and drafting. Numerical and editorial claims were checked against independent formulations and preserved computational evidence. Responsibility for the work rests with the named author.

13 Conclusions

Side-specific interval and directional disjoint-group penalties admit explicit original-row proximal calculations in the stated positive diagonal-metric setting. The box-contraction argument, scalar threshold and joint dual budget identify the mathematical conditions needed to recover the intended original objective. Equivalent lifting, directional row splitting and compact epigraphs delimit the claim: these objectives were already expressible, and no priority assertion follows from retaining the original principal rows.

The completed computational predictions are negative. The original 456-instance study has a primary 168-observation median reduction of 0.499839%, with 95% limits -3.291967% and 3.944835%. The separate grouped 210-instance study has a 144-member primary score of -212.449875%, limits -225.336618% and -192.873759%, and 9 retained failed pairs. Neither establishes the registered 10% gain.

Correct local mathematics and original-space agreement therefore do not by themselves establish a faster complete numerical operation. Future computational work should target a specific new mechanism and evaluate it prospectively against the strongest compatible references. The present findings support restricted derivations, reproducible implementation analysis and complete negative evidence; they do not establish operational safety, field benefit or general solver superiority.

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